Getting Started with Python

Setup Guide and Tutorial

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January 2026

Welcome!

This tutorial will guide you through setting up your Python environment for the course.

What you'll learn:

- How to access course materials (slides & Jupyter notebooks)
- How to set up your coding environment
- How to run Python code for assignments
- How to organize and submit your project

Prerequisites:

- A computer with internet access
- Basic familiarity with file systems
- Willingness to learn!

Course Materials

All course materials are hosted on GitHub:

Repository URL

`https://github.com/matteogarbelli/NMMF`

What's in the repository:

- **Lecture slides** (PDF and LaTeX source files)
- **Jupyter notebooks** (.ipynb) with code examples
- **Python scripts** for simulations and visualizations
- **Project templates** for your assignments
- **Data files** for exercises

Tutorial Outline

Step 1: Create a GitHub Account

Step 2: Install GitHub Desktop

Step 3: Clone the Course Repository

Step 4: Choose Your Python Environment
- Option A: Using Google Colab
- Option B: Local Setup with VS Code

Step 5: Install Required Python Packages

Why GitHub?

GitHub is a platform for version control and collaboration.

Benefits for this course:

- Access all course materials in one place
- Keep your code organized and backed up
- Track changes to your project over time
- Easy sharing and collaboration
- Industry-standard tool for data science and software development

Note: GitHub is free for students and widely used in academia and industry!

Creating Your GitHub Account

1. Go to `https://github.com`
2. Click **"Sign up"** in the top-right corner
3. Enter your details:
   - Use your university email (you may get student benefits!)
   - Choose a professional username
   - Create a strong password
4. Verify your email address
5. **Optional:** Apply for GitHub Student Developer Pack at `https://education.github.com/pack`

Tutorial Outline

Why GitHub Desktop?

GitHub Desktop provides a user-friendly interface to work with Git repositories.

Advantages:

- No command-line knowledge required
- Visual interface for commits
- Easy to sync with GitHub
- Cross-platform (Windows, Mac, Linux)

Alternative:

- You can also use Git command line
- Or download as ZIP (but no version control)

Recommended: GitHub Desktop for beginners!

Installing GitHub Desktop

1. Go to `https://desktop.github.com`
2. Click **"Download for [Your OS]"**
3. Run the installer and follow the installation wizard
4. Launch GitHub Desktop
5. Sign in with your GitHub account
6. Configure your name and email for commits

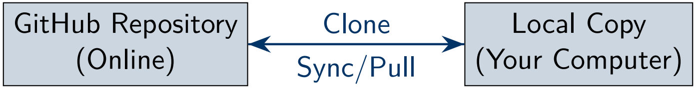
Note

These settings will be used to identify your contributions in the repository.

Tutorial Outline

What is "Cloning"?

Cloning creates a local copy of the repository on your computer.



Benefits:

- Work offline
- Make changes without affecting the original
- Get updates when new materials are added

Cloning the Repository with GitHub Desktop

1. Open GitHub Desktop
2. Click **File** $\rightarrow$ **Clone Repository**
3. Select the **URL** tab
4. Paste the repository URL:

   `https://github.com/matteogarbelli/NMMF`

5. Choose a local path (e.g., `Documents/NMMF`)
6. Click **Clone**

Wait for the download to complete!

Keeping Your Repository Up-to-Date

As new materials are added throughout the course, you'll need to update your local copy.

To get the latest updates:

1. Open GitHub Desktop
2. Select the NMMF repository from the left sidebar
3. Click **Fetch origin** (top-right)
4. If updates are available, click **Pull origin**

Recommendation

Check for updates before each lecture!

Tutorial Outline

Two Options for Running Python Code

You have two main options for running the Jupyter notebooks:

Option A: Google Colab

+ No installation required
+ Free GPU access
+ Works on any computer
+ Auto-saves to Google Drive
- Requires internet connection
- Session timeouts

Best for: Beginners, quick testing

Option B: Local Setup (VS Code)

+ Full control over environment
+ Works offline
+ Better for large projects
+ Professional workflow
- Requires installation
- Setup time needed

Best for: Serious development

You can use both! Start with Colab, then move to local setup.

Getting Started with Google Colab

Google Colab is a free cloud-based Jupyter notebook environment.

Setup Steps:

1. Go to `https://colab.research.google.com`
2. Sign in with your Google account
3. Click **File** $\rightarrow$ **Upload notebook**
4. Navigate to your cloned NMMF folder
5. Upload the .ipynb file you want to work with
6. Start coding!

Connecting Colab to GitHub (Advanced)

You can open notebooks directly from GitHub in Colab!

Method 1: Using Colab's GitHub Integration

1. In Colab, click **File** → **Open notebook**
2. Select the **GitHub** tab
3. Enter: `matteogarbelli/NMMF`
4. Choose your notebook

Method 2: Direct URL

Replace `github.com` with `colab.research.google.com/github` in any notebook URL:

```
colab.research.google.com/github/matteogarbelli/NMMF/blob/main/notebook.
    ipynb
```

Working with Colab

Key Features:

- **Cells:** Code cells (Python) and text cells (Markdown)
- **Running code:** Click play button or press `Shift + Enter`
- **Installing packages:**
  - Most packages (NumPy, Pandas, Matplotlib) are pre-installed
  - For others: `!pip install package_name`
- **Saving work:**
  - Auto-saves to Google Drive
  - Download: **File** → **Download** → **Download .ipynb**
- **Runtime:** Free tier gives 12-hour sessions with GPU/TPU access

Why Visual Studio Code?

VS Code is a powerful, free code editor with excellent Python support.

Features:

- Built-in Jupyter notebook support
- Intelligent code completion (IntelliSense)
- Integrated terminal
- Git integration
- Extensions for Python, LaTeX, and more
- Cross-platform (Windows, Mac, Linux)

Alternative Editors:

- JupyterLab (browser-based, like Colab but local)
- PyCharm (more advanced, heavier)
- Spyder (MATLAB-like interface)

Step 1: Install Python

Before installing VS Code, you need Python on your system.

Install Python from python.org

1. Go to `https://www.python.org/downloads/`
2. Download the latest Python 3 version (3.10 or newer)
3. Run the installer
   - **Important:** Check "Add Python to PATH" during installation
   - Choose default options for everything else
4. Verify installation: Open terminal/command prompt and type:
   - `python --version`
   - Should display Python 3.x.x

Note: We'll install the required packages (NumPy, Matplotlib, etc.) separately using pip.

Step 2: Install Visual Studio Code

1. Go to `https://code.visualstudio.com`
2. Click **Download** for your operating system
3. Run the installer
4. Launch VS Code
5. Install the **Python extension**:
   - Click the Extensions icon (left sidebar) or press `Ctrl+Shift+X`
   - Search for "Python"
   - Install the extension by Microsoft
6. **Optional:** Install the **Jupyter extension** for better notebook support

Step 3: Open Your Repository in VS Code

1. Open VS Code
2. Click **File** → **Open Folder**
3. Navigate to your cloned NMMF folder
4. Click **Select Folder**
5. You should see the repository structure in the Explorer panel (left sidebar)
6. Open any `.ipynb` file to start working!

First time? VS Code may prompt you to:

- Select a Python interpreter (choose the Python 3.x you installed)
- Install additional packages (ipykernel) - click "Install"

Working with Jupyter Notebooks in VS Code

Running notebooks:

- Open any `.ipynb` file
- Select kernel (Python 3.x) in top-right corner
- Run cells individually or all at once
- Use `Shift + Enter` to run current cell

Useful shortcuts:

- `Ctrl + Enter`: Run cell without moving to next
- `Shift + Enter`: Run cell and move to next
- `A`: Add cell above (in command mode)
- `B`: Add cell below (in command mode)
- `DD`: Delete cell (in command mode)

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Essential Python Libraries for This Course

Installing required packages:
Open a terminal (Command Prompt on Windows, Terminal on Mac/Linux) and run:

```
pip install numpy scipy matplotlib pandas jupyter
```

Additional useful packages:

```
pip install seaborn scikit-learn ipykernel
```

Verify installation: Open Python and try:

```
import numpy as np
import matplotlib.pyplot as plt
print("Success!")
```

Tip: The `ipykernel` package is needed for Jupyter notebooks in VS Code.

Managing Packages with pip

pip is Python's package manager.

Useful commands:

```
# List installed packages
pip list

# Install a package
pip install package_name

# Update a package
pip install --upgrade package_name

# Install specific version
pip install package_name==1.2.3

# Uninstall a package
pip uninstall package_name
```

Creating Your Project

For the final project, you'll need to:

1. Choose a topic from the provided list
2. Create a new folder in your local repository:
   - `NMMF/my_project/`
3. Organize your project:
   - `notebook.ipynb` - Your main code and analysis
   - `report.pdf` - Written report (optional: use LaTeX)
   - `data/` - Any data files you use
   - `figures/` - Plots and visualizations
   - `README.md` - Project description
4. **Submission:**
   - Create your own GitHub repository for the project
   - Or compress your project folder and submit via email

For any questions

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Repository: https://github.com/matteogarbelli/NMMF